

HANOVER COUNTY MUNICIPAL, VA, USA

WMO: 723084

Lat: 37.708N

Lon: 77.434W

Elev: 63

StdP: 100.58

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93775

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -7.0	(d) -18.7	(e) 0.7	(f) -5.1	(g) -16.2	(h) 0.9	(i) -3.4	(j) 9.3	(k) 10.1	(l) 8.4	(m) 8.6	(n) 1.3	(o) 290	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.1	(c) 35.0	(d) 24.1	(e) 33.5	(f) 23.7	(g) 32.3	(h) 23.4	(i) 25.7	(j) 31.8	(k) 25.1	(l) 31.0	(m) 24.5	(n) 29.9	(o) 3.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.1	27.6	23.5	18.4	27.1	22.9	17.7	26.7	79.2	32.1	76.5	30.9	74.2	29.9	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.0	(b) 6.9	(c) 5.7	(e) -13.8	(f) 36.7	(g) 3.2	(h) 1.8	(i) -16.0	(j) 38.0	(k) -17.9	(l) 39.1	(m) -19.6	(n) 40.1	(o) -21.9	(p) 41.4		
(a) 8.0	(b) 6.9	(c) 5.7	(e) -14.5	(f) 27.4	(g) 3.0	(h) 1.0	(i) -16.6	(j) 28.1	(k) -18.3	(l) 28.8	(m) -20.0	(n) 29.3	(o) -22.2	(p) 30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	3.0	4.7	8.9	14.6	19.3	23.7	25.8	25.0	21.3	15.3	9.5	5.2
	DBStd	9.20	5.98	5.43	5.55	5.00	4.34	3.29	2.75	2.73	3.55	4.67	4.96	5.15
	HDD10.0	743	230	166	91	13	0	0	0	0	0	8	68	168
	HDD18.3	2141	475	383	296	135	43	2	0	0	13	118	267	408
	CDD10.0	2476	13	17	56	150	289	411	489	466	340	172	54	18
	CDD18.3	831	0	1	3	22	73	163	231	208	103	24	2	1
	CDH23.3	7967	1	4	42	262	696	1601	2370	1960	818	198	13	3
	CDH26.7	3224	0	0	6	75	224	661	1095	843	273	47	1	0
Wind	WSAvg	2.1	2.4	2.5	2.7	2.7	2.2	2.0	1.8	1.6	1.6	1.8	2.0	2.1
Precipitation	PrecAvg	1107	83	70	102	84	98	102	110	107	108	79	73	92
	PrecMax	1629	175	161	216	205	246	195	211	361	380	168	253	216
	PrecMin	843	31	22	6	31	26	26	30	36	0	0	0	24
	PrecStd	181	33	34	45	37	50	46	44	63	81	45	54	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.4	27.4	31.3	32.8	36.0	37.3	36.3	33.9	30.7	25.2	22.8
		MCWB	15.8	15.9	16.5	19.3	22.5	23.7	24.6	24.4	22.9	21.8	18.2	16.7
	2%	DB	18.8	20.0	24.0	28.8	31.0	33.7	35.2	34.2	31.8	27.7	22.5	19.5
		MCWB	14.3	14.0	15.5	18.1	21.7	23.7	24.4	23.9	22.7	20.1	16.2	16.0
	5%	DB	16.3	17.3	21.6	26.7	29.1	32.3	33.7	32.8	29.7	25.6	20.7	17.1
		MCWB	12.3	11.4	14.0	17.3	21.1	23.4	24.0	23.7	21.7	19.4	15.8	13.3
	10%	DB	12.8	14.1	18.8	23.8	27.5	30.7	32.2	31.3	27.8	23.2	18.0	13.9
		MCWB	8.7	9.3	12.6	16.3	20.1	22.6	23.8	23.2	21.2	18.3	14.0	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	17.8	18.3	21.1	24.2	25.8	26.8	26.4	25.4	23.9	20.0	19.2
		MCDB	19.9	21.5	23.7	27.1	29.6	31.2	32.3	33.0	29.6	27.0	22.6	20.9
	2%	WB	15.3	15.2	16.8	19.7	22.9	24.9	25.8	25.6	24.3	22.3	18.4	16.4
		MCDB	18.3	18.9	22.2	26.2	28.6	30.8	32.2	31.8	28.5	25.9	21.1	18.6
	5%	WB	12.8	12.6	15.4	18.5	22.1	24.2	25.2	24.9	23.5	20.8	16.7	14.0
		MCDB	15.5	15.9	20.0	24.5	27.5	30.1	31.4	30.5	27.2	24.3	19.3	16.3
	10%	WB	9.1	9.8	13.6	17.3	21.1	23.6	24.6	24.2	22.7	19.1	14.8	10.9
		MCDB	12.1	14.0	17.9	22.5	26.0	28.9	30.2	28.9	25.9	22.2	17.9	13.1
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.6	12.4	13.0	11.9	11.4	11.1	11.0	11.1	11.7	11.9	10.8
		MCDBR	14.0	15.2	15.5	15.6	13.3	13.0	12.9	12.7	13.0	13.5	13.9	13.6
		MCWBR	10.4	10.2	8.7	7.0	6.0	5.1	4.5	4.3	5.2	6.9	8.4	10.1
	5% WB	MCDBR	12.9	13.6	13.6	13.2	11.5	11.4	11.2	10.6	10.0	10.9	11.0	12.0
		MCWBR	11.1	10.7	9.2	6.8	5.6	4.9	4.5	4.3	4.8	6.5	7.9	10.3
Clear-Sky Solar Irradiance	taub	0.300	0.314	0.347	0.389	0.427	0.485	0.503	0.505	0.408	0.352	0.319	0.304	
	taud	2.529	2.477	2.426	2.322	2.242	2.124	2.097	2.073	2.330	2.485	2.535	2.553	
	Ebn at Noon	893	921	916	890	859	806	787	775	841	864	861	863	
	Edh at Noon	77	92	107	125	138	156	159	159	116	91	76	71	
All-Sky Solar Radiation	RadAvg	2.28	3.05	4.04	5.08	5.76	6.28	6.01	5.42	4.42	3.44	2.57	2.01	
	RadStd	0.19	0.34	0.32	0.39	0.55	0.44	0.31	0.26	0.47	0.41	0.20	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (74 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+84

Nomenclature: See separate page